

- [34] R. Zellers, A. Holtzman, H. Rashkin, Y. Bisk, A. Farhadi, F. Roesner, and Y. Choi. Defending against neural fake news. *Advances in neural information processing systems*, 32, 2019.
- [35] H. Zhang, B. L. Edelman, D. Francati, D. Venturi, G. Ateniese, and B. Barak. Watermarks in the sand: Impossibility of strong watermarking for generative models. *arXiv preprint arXiv:2311.04378*, 2023.
- [36] X. Zhao, P. Ananth, L. Li, and Y.-X. Wang. Provable robust watermarking for ai-generated text. *arXiv preprint arXiv:2306.17439*, 2023.
- [37] W. Zhou, Y. E. Jiang, E. Wilcox, R. Cotterell, and M. Sachan. Controlled text generation with natural language instructions. In *International Conference on Machine Learning*, pages 42602–42613. PMLR, 2023.

A Comparison with Watermark Stealing

In Section 4, we demonstrated the effectiveness of adversarial paraphrasing by comparing it against both simple and recursive paraphrasing. In this section, we extend our evaluation by comparing adversarial paraphrasing with the Watermark Stealing attack [12], a targeted approach specifically designed to compromise watermarking methods—unlike our more general-purpose (universal) attack.

Following the experimental setup proposed by Jovanović *et al.* [12], we use the LLaMA2-7B-Chat model [31], watermarked using the KGW scheme introduced by Kirchenbauer *et al.* [14], to generate a dataset of 2000 watermarked samples. Consistent with our prior experiments, the model is provided with the first 20 words of each of the 2000 AI-generated texts from the MAGE dataset [18] as a prefix and generates approximately 200 to 600 tokens per sample conditioned on that context. The watermarking parameters match those used in the original watermark stealing study [12]. Using the learned watermarking scheme, we then perform a scrubbing attack as described in the same study, employing both Mistral-7B [11] and LLaMA2-7B [31] as paraphraser.

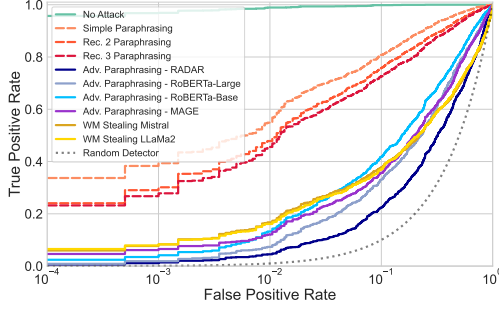


Figure 6: ROC curves illustrating the AI text detection performance of KGW watermark under different attacks, including simple paraphrasing, recursive paraphrasing, watermark stealing, and adversarial paraphrasing. The false positive rate (FPR) axes is displayed in log-scale to highlight fine-grained distinctions in the low-FPR region. It can be seen that adversarial paraphrasing outperforms all baselines, including watermark stealing, in pushing the detector’s performance closer to that of a random one.

Figure 6 presents the ROC curves for KGW watermark detection performance under various attack scenarios, including simple paraphrasing, recursive paraphrasing, watermark stealing, and adversarial paraphrasing. From the ROC curves,

it can be observed that while adversarial paraphrasing guided by certain detectors demonstrates slightly reduced effectiveness compared to watermark stealing in high false positive rate (FPR) regions, it consistently outperforms all baselines in degrading detector performance in the low FPR regime ($FPR \leq 1\%$). Notably, RADAR [8] proves to be the most effective guidance detector in this setting. Table 7 reports the same three key metrics introduced in our Experiments section: the Area Under the ROC Curve (AUC), the True Positive Rate at 1% False Positive Rate ($T@1\%F$), and GPT-4o’s automated quality ratings (Rating). The results

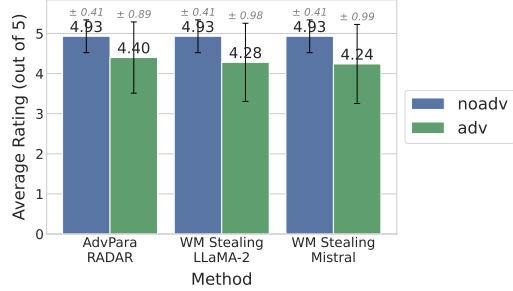


Figure 7: Text quality evaluations using GPT-4o, comparing watermark stealing and adversarial paraphrasing (guided by RADAR [8], which demonstrated the highest attack effectiveness in this case) against simple paraphrasing. The results show that adversarial paraphrasing produces higher-quality text than watermark stealing.

	KGW WM		Rating
	AUC (↓)	T@1%F (↓)	
No Attack	0.999	0.989	—
Simple Paraphrase	0.930	0.549	4.93 ± 0.41
Rec. Para. 2	0.903	0.478	4.95 ± 0.22
Rec. Para. 3	0.889	0.455	4.92 ± 0.34
WM Stealing (Mistral-7B)	0.669	0.167	4.24 ± 0.99
WM Stealing (LLaMA2-7B)	0.670	0.161	4.28 ± 0.98
AdvPara (RoBERTa-Large)	0.703	0.073	4.62 ± 0.76
AdvPara (RoBERTa-Base)	0.751	0.132	4.76 ± 0.53
AdvPara (MAGE)	0.707	0.121	4.84 ± 0.46
AdvPara (RADAR)	0.619	0.045	4.40 ± 0.89

Table 7: Detection performance of KGW Watermark in distinguishing between AI-generated and human-written text under different attack scenarios. Metrics reported include AUC and TPR at 1% FPR. Additionally, we present the mean $\pm$ standard deviation of quality ratings given by GPT-4o. It can be observed that adversarial paraphrasing results in the lowest AUC and TPR@1%FPR after the attack.

indicate that adversarial paraphrasing guided by RADAR yields the lowest AUC and T@1%F values post-attack. Figure 7 further details the text quality assessments for the watermark stealing and adversarial paraphrasing attacks, compared against simple paraphrasing. It shows that adversarial paraphrasing results in better text quality compared to watermark stealing.

B Text Token Counts in Efficiency Assessment

The exact runtime per sample may vary depending on the sequence length, which is inherently determined by the length of the input text. To ensure a fair comparison—such that adversarial paraphrasing generates a comparable number of tokens to simple paraphrasing—we report the mean and standard deviation of token counts in Table 8. As shown, both simple paraphrasing and adversarial paraphrasing with different guidance detectors produce a similar mean number of tokens. However, adversarial paraphrasing exhibits a larger variance in token counts, reflecting greater variability in output length.

Text	Mean Token Count	Std Dev
Original texts	173.73	38.46
Simple Paraphrase	170.81	39.69
AdvPara (RoBERTa-Base)	175.30	51.27
AdvPara (RoBERTa-Large)	171.25	45.39
AdvPara (RADAR)	169.68	60.07
AdvPara (MAGE)	164.18	54.39

Table 8: Mean and standard deviation of token counts for the original, simple paraphrased, and adversarially paraphrased texts used in our efficiency evaluation. All token counts were obtained using the LLaMA-3 tokenizer.

C Detailed Token Statistics for Evaluated Datasets

We report the detailed token statistics for all datasets used in our main experiments in Table 9. The token counts are obtained from the LLaMA-3 tokenizer.

Dataset	Min # tokens	Max # tokens	Mean # tokens
MAGE human texts	110	305	~179
MAGE AI texts	110	525	~175
KGW watermarked texts	199	602	~269
Unigram watermarked texts	161	602	~319

Table 9: Token statistics for the texts used in our evaluation, obtained from the LLaMA-3 tokenizer.

D What Happens if Detector Guidance is Not Applied at All Decoding Steps?

In our default configuration, detector guidance is applied at every iteration of adversarial paraphrasing (Figure 1). This setup achieves high attack effectiveness (Section 4.2) while maintaining acceptable latency (Section 4.3). In this section, we conduct an ablation study to investigate the impact of applying detector guidance less frequently. Specifically, guidance is applied once every N steps during the auto-regressive generation process of the paraphraser. In this setting, the paraphraser generates $N - 1$ tokens in a standard (non-adversarial) manner, and the N -th token is then sampled with detector guidance. For the guidance detector, we employ OpenAI-RoBERTa-Large and perform adversarial paraphrasing on 500 randomly selected text samples. The resulting mean AUC and TPR@1%FPR across all eight deployed detectors are reported in Table 10.

It can be observed that applying detector guidance at every decoding step yields the highest attack effectiveness, while performance gradually decreases as N increases. Nevertheless, even with $N = 5$, adversarial paraphrasing maintains superior attack effectiveness compared to the simple paraphrasing baseline. Moreover, using a larger N reduces attack latency, as fewer detector-guided steps are required.